

SoS 2026

CS20 - Generative and Agentic AI

Ayush Sharma (24B2475)

Mentor: Naman Tunwal

Timeline: May – July 2026

Project Overview

This project focuses on the engineering, benchmarking, and optimisation of cloud-accelerated, cost-free autonomous agentic workflows applied to enterprise market analytics. Moving away from localised hardware bottlenecks and commercial costs, the core execution pipeline synthesises high-volume, unstructured corporate filings with real-time web-scraping data through a multi-agent LangGraph architecture.

Rather than generating open-ended conversational text, the system terminates in a strict verification loop to produce two explicit, deterministic deliverables:

1. **The Human-Readable Artifact (M&A_Diligence_Brief.md):** A strictly formatted Markdown executive report featuring a target overview, an automated financial claim-vs-reality discrepancy matrix, and cross-referenced risk flags.
2. **The Machine-Readable Artifact (Financial_Risk_Payload.json):** A Pydantic-enforced, structured data object mapping verified financial metrics and documented risk citations, engineered for direct injection into downstream enterprise financial modelling pipelines.

Timeline

Week 1 (May 26 – Jun 1) | Foundations of LLMs & Strategic Prompting

- **Objectives:** Investigate the operational landscape of Large Language Models (LLMs) and core distribution semantics. Study deterministic vs. probabilistic generation constraints.
- **Implementation:** Write test functions to evaluate how Gemini 2.5 Flash and Llama 3.3 70B (using Groq) handle **Few-shot prompting** and **Chain-of-Thought (CoT)** reasoning when extracting qualitative metrics (e.g., SWOT points) from an unformatted block of financial text.

Week 2 (Jun 2- Jun 8) | Transformer Architecture & Cloud API Engine Setup

- **Objectives:** Study the mathematical foundations of the Transformer architecture, focusing on multi-head self-attention mechanics and positional encodings.
- **Implementation:** Analyse tokenisation and context window processing. Set up a programmatic cloud development environment **after Week 1** by integrating the **Google AI Studio (Gemini)** and **Groq Cloud** developer interfaces to deploy zero-cost, high-throughput model endpoints for complex long-context reasoning.

Week 3 (Jun 9 – Jun 15) | High-Performance Document Processing & Local Retrieval

- **Objectives:** Explore strategic document ingestion methods, including **recursive character chunking**, structural parsing of financial reports, and **retrieval/reranking basics**.
- **Implementation:** Configure localised vector stores (ChromaDB) to manage enterprise documents and corporate reports. Implement a multi-stage retrieval pipeline paired with an open-source **Cross-Encoder reranker** to pass highly dense context packages to the cloud inference layer.

Week 4 (Jun 16 – Jun 22) | Orchestration Frameworks & High-Context RAG Pipeline

- **Objectives:** Study high-level abstraction frameworks for chaining LLM processes and managing stateful conversational interactions.
- **Implementation:** Construct a production-style, high-context Retrieval-Augmented Generation (RAG) pipeline utilising Gemini's extended context window to synthesise dense corporate filings and historical data charts alongside conversation memory.
- **Milestone: Mid-term Report Submission (3rd Week of June / End of Week 4)**

Week 5 (Jun 23 – Jun 29) | Graph-Based Multi-Agent Architectures & State Machines

- **Objectives:** Transition from static chains to autonomous multi-agent networks. Study the ReAct paradigm and implement cyclic, graph-based execution loops.
- **Implementation:** Build state-machine routing environments using LangGraph. Force separate model instances to document logic blocks (Thought -> Action -> Observation) while maintaining a shared state memory across multiple turns.

Week 6 (Jun 30– Jul 6) | Tool Integration & Deterministic Boundaries

- **Objectives:** Interface the language model's reasoning engine with deterministic external systems and ensure reliable, structured data extraction.
- **Implementation:** Construct executable Python interfaces (live web-news scraper, local ChromaDB financial index retriever) and bind them as node tools in LangGraph. Enforce Pydantic validation to ensure the LLM returns structured JSON data for programmatic use.

Week 7 (Jul 7 – July 13) | Capstone: Autonomous Market Intelligence Engine

- **Objectives:** Finalise framework-level agentic patterns, focusing on automated deal flow, reflection boundaries, and error recovery within cloud-orchestrated networks.
- **Implementation:** Finalise the M&A Due Diligence Engine. Implement a strict "Reflection & Critique" loop where an Auditor Agent extracts corporate assertions, triggers the Scraper tool for real-time market data, self-corrects contradictions, and compiles both a structured data schema and a polished executive threat report.

Week 8 (Jul 14 – Jul 20) | Quantitative System Evaluation & Final Synthesis

- **Objectives:** Investigate alignment drift and prompt fragility within asynchronous loops. Quantify the performance trade-offs between static high-context RAG and dynamic agentic execution.

- **Implementation:** Deploy LangSmith or Ragas to run programmatic evaluations on the Capstone engine. Measure and score the system on Context Precision, Faithfulness, and Tool-calling accuracy.
- **Milestone:** End-term Report Submission.

Core References

Technical Papers & Literature

- Vaswani et al. *Attention Is All You Need* (Fundamental Transformer Architecture).
- Yao et al. *ReAct: Synergizing Reasoning and Acting in Language Models* (ICLR).
- Lilian Weng. *LLM Powered Autonomous Agents* (Open-source Technical Analysis).
- Anthropic Research. *Building Effective Agents* (Architectural Design Patterns).

Instructional Lectures & Courses

- Karpathy, A. *Intro to Large Language Models & Deep Dive into LLMs* (Neural Mechanics series).
- Hugging Face Academy. *The Hugging Face Agents Course* (Units 0-2).
- DeepLearning.AI. *Prompt Engineering & Framework Abstractions for Developers*.

Documentation Frameworks

- *LangChain Documentation* (Core Abstractions, Tool Interfaces, and State Memory).
- *ChromaDB API Specification* (Vector Indexing & Collection Management).
- *Google AI Studio & Groq Cloud Developer API Specifications* (Cloud Model Infrastructures).